

Hardware Module

Components Used

- ESP32 Microcontroller
- Load Cell Sensor
- HX711 Load Cell Amplifier
- Vibration Motor
- Rechargeable Battery
- Bluetooth Communication Module (Built into ESP32)
- Connecting Wires
- Crutch Frame

Hardware Working

The Smart Crutch measures the weight applied by the user using a Load Cell sensor. The sensor data is amplified through the HX711 module and processed by the ESP32 microcontroller.

The ESP32 analyzes the load data and provides real-time feedback through a vibration motor whenever the applied weight exceeds the safe rehabilitation limit.

The processed data is transmitted wirelessly to a mobile application through Bluetooth for monitoring and progress tracking.

Objectives

- Monitor weight bearing during rehabilitation.
- Detect improper load distribution.
- Provide real-time vibration feedback.
- Support remote rehabilitation monitoring.
- Improve recovery safety and effectiveness.

Future Hardware Enhancements

- GPS Module
- GSM Emergency Alert System
- Fall Detection Sensor
- Advanced Gait Analysis Sensors
- Cloud Connectivity